

Toll-Like Receptors 2 and 4 Are Involved in the Induction of Graft-Versus-Host-Disease in Mice.

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Abstract

BACKGROUND: Intestinal Graft-versus-Host disease is a frequent and often lethal complication after allogeneic stem cell transplantation. Since NOD2 polymorphisms have been recognized as potential triggers of severe intestinal GvHD in humans, we have developed murine transplantation models to investigate the role of different pattern recognition receptors (PRR) in GvHD and GvL. Here we report our results on the role of TLR2 and TLR4 for the induction of GvHD.

METHODS: Severity of GvHD in wildtype (wt) C57B/10 (H-2Db), TLR2^{-/-}, TLR4^{-/-}, and combined TLR2^{-/-}TLR4^{-/-} C57B/10 mice was investigated. Mice received treosulfan 2000 mg/kg from day -3 to -1 and cyclophosphamide 200 mg/kg day -1 prior to injection of 10×10⁶ H-2Dd BM cells and 5×10⁶ splenocytes (SC). Survival and GvHD score were assessed daily. Engraftment was determined every 2 weeks in pB and at the end of the experiments in bone marrow by flow cytometry. T cell alloreactivity in GvH direction was assessed by MLR using splenocytes as stimulators from PRR-deficient mice or wt as control and CFSE-staining as read-out. The relevance of PRR ligands for the enhancement of GvH alloreactivity was determined by addition of lipid A, lipopeptides, or CpG.

RESULTS:

in vivo data: The transfer of 10×10⁶ BMC + 5×10⁶ SC induced a severe GvHD in all wt recipients, leading to death of 90% of the animals within 20 days. Recipient mice lacking either TLR2 or TLR4 showed only a slightly and not significantly decreased GvHD lethality. In recipients lacking both PPRs, i.e.

TLR2 and TLR4, GvHD was generally milder and the majority (60%) of the animals survived until day 20 ($p<0.05$). However, the long term survival was not significantly improved. Differences in clinical severity of GvHD were confirmed histologically. In vitro data: Stimulation with cells from TLR2 $-/-$ and TLR4 $-/-$ mice resulted in a decreased alloreactivity in MLR. A median of 2% of Balb/c CD4 $+$ T cells proliferated in response to C57B/10 stimulators. The addition of the TLR2 and TLR4 ligands lipopeptide, Lipid A and CpG significantly ($p<0.05$) increased the proliferation of CD4 $+$ T cells in a specific manner more than twofold.

CONCLUSION: Our in vivo and in vitro data consistently show that bacterial components are involved in triggering GvH alloreactivity via different types of PPRs. Binding of bacterial substances to TLR2 and TLR4 leads to activation of the immune system and subsequent induction of GvHD. Our data provide an experimental basis for the development of strategies for modulation of the intestinal gut flora by selective gut decontamination and/or probiotic regimens to prevent GvHD in humans.

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Author notes

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